

KCPC Week 2 Solutions

1 Wishful Thinking

Basic: Complete The Square

Subtract the right-hand side from the left:

$$x^2 + y^2 - \frac{(x+y)^2}{2} = \frac{(x-y)^2}{2} \geq 0.$$

Equality holds exactly when $x = y$.

Key insight: Wish for a square because squares are automatically non-negative.

Easy: Make The Sum Telescope

Use partial fractions:

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

Therefore all intermediate terms cancel:

$$\sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{n+1} = \frac{n}{n+1}.$$

Key insight: Rewriting each term as a difference exposes cancellation.

Medium: A Number Puzzle

Set $x = u + v$ and $y = u - v$. The equation becomes $u^2 + v^2 = z^2$. Take the Pythagorean triple

$$u = m^2 - 1, \quad v = 2m, \quad z = m^2 + 1$$

for any integer $m \geq 3$. Then

$$x = m^2 + 2m - 1, \quad y = m^2 - 2m - 1, \quad z = m^2 + 1$$

is positive and satisfies the original equation. Varying m gives infinitely many solutions; moreover, $x - y = 4m \neq 0$.

Key insight: The substitution turns the desired equation into a familiar one.

Easy: Improve The Recurrence

Define $b_n = a_n + 1$. Then

$$b_{n+1} = a_{n+1} + 1 = 2a_n + 2 = 2b_n, \quad b_1 = 2.$$

Hence $b_n = 2^n$, so

$$a_n = 2^n - 1.$$

Key insight: Changing variables absorbs the awkward constant term.

Very Hard: Factor The Equation

Multiplying by nxy gives $n(x + y) = xy$. Rearranging and adding n^2 to both sides gives

$$(x - n)(y - n) = n^2.$$

Both x and y must exceed n . Thus, for every positive divisor d of n^2 , set

$$x = n + d, \quad y = n + \frac{n^2}{d}.$$

These are all ordered positive solutions, since every factorisation of n^2 arises this way.

Key insight: Adding the right term changes a sum equation into a factorisation.

2 Symmetry

Basic: Pairing Game

Number the rows $1, \dots, 2r$. Reflect every first-player move across the horizontal line between rows r and $r + 1$: row i is paired with row $2r + 1 - i$, using the same columns. No row is paired with itself.

Initially the board is symmetric and empty. Suppose it is symmetric after a second-player reply. If the first player now plays a legal domino, its reflected cells were also empty: an occupied reflected cell would, by symmetry, imply an occupied cell in the first player's move. The reflected domino lies in a different row, so the two new dominoes are disjoint. The reply is therefore legal and restores symmetry. This proves the invariant after every round. Since the board is finite, play terminates, and the second player always has a reply whenever the first player can move. The first player must run out of moves first.

Key insight: Pair rows with no fixed row, and prove the mirrored response is always legal.

Easy: Coins On A Circular Table

The first player places a coin at the centre. After every later move, the first player places a coin in the position obtained by rotating the opponent's coin by 180° . The table remains symmetric after each response, so the mirrored placement is legal whenever the opponent's placement is legal. More explicitly, if the coin radius is ρ , a legal opponent

coin has centre at distance at least 2ρ from the central coin's centre. Its reflected centre is therefore at distance at least 4ρ from it, so the two coins do not overlap. Any overlap with an earlier coin would reflect to an overlap in the opponent's move, which was legal. The finite table can hold only finitely many coins, so the first player moves last. As usual, assume at least one coin fits on the table.

Key insight: Occupy the unique fixed point, then pair all remaining positions.

Medium: A Symmetric Integral

Let $I = \int_0^1 f(x) dx$. Substituting $u = 1 - x$ gives

$$I = \int_0^1 f(1 - x) dx.$$

Adding the two expressions and using the given identity,

$$2I = \int_0^1 (f(x) + f(1 - x)) dx = \int_0^1 1 dx = 1.$$

Hence $I = \frac{1}{2}$.

Key insight: Reflecting the interval pairs every value with its complement.

Hard: Odd Skew-Symmetric Matrices

Using invariance of the determinant under transposition,

$$\det A = \det A^T = \det(-A) = (-1)^n \det A.$$

When n is odd this says $\det A = -\det A$, so $2\det A = 0$ and $\det A = 0$.

Key insight: Transpose symmetry turns the determinant into its own negative.

Very Hard: Six People

Represent people by vertices and colour each edge red for acquaintances and blue for strangers. Choose a vertex v . At least three of its five incident edges have one colour; by symmetry suppose va, vb, vc are red. If one of ab, bc, ca is red, it forms a red triangle with v . Otherwise ab, bc, ca are all blue and form a blue triangle.

Key insight: Colour symmetry reduces two relationship cases to one.

Stretch: Chomp: Poisoned Chocolate

Suppose the second player has a winning strategy. If the first player eats only the top-right square, this strategy specifies a reply at some unpoisoned square q . The first player could instead have chosen q as their opening move: every move at q also eats the top-right square, so the resulting board is exactly the board after those two hypothetical moves. It is now the second player's turn, whereas in the hypothetical game it was the

first player's turn. The first player can therefore take over the supposed winning *second-player* strategy, a contradiction. Thus the first player has a winning strategy, although this argument does not specify their first move.

Key insight: A hypothetical winning reply can be stolen as the first move.

3 Extreme Principle

Basic: A Finite Set Closed Under Addition

Let $M = \max S$ and $m = \min S$, which exist because S is finite and non-empty. If $M > 0$, closure gives $2M \in S$ with $2M > M$, contradicting maximality. Thus $M \leq 0$. If $m < 0$, closure gives $2m \in S$ with $2m < m$, contradicting minimality. Thus $m \geq 0$. Every element of S lies between m and M , so $m = M = 0$ and $S = \{0\}$. Conversely, $\{0\}$ is indeed closed under addition.

Key insight: The maximum excludes positive elements; the minimum excludes negative ones.

Easy: Minimum Degree Forces A Cycle

Choose a longest path $v_0 v_1 \dots v_k$. Every neighbour of v_k must already lie on the path, since a new neighbour would extend it. Because v_k has at least two neighbours, it is adjacent to some v_i with $i < k - 1$. The edge $v_k v_i$ together with the path from v_i to v_k forms a cycle.

Key insight: An endpoint of a longest path cannot escape to a new vertex.

Medium: An Empty Triangle

Among all non-degenerate triangles whose vertices belong to the set, choose one of minimum positive area. If another given point lay in its interior, joining that point to the three vertices would split the triangle into three smaller triangles of positive area. At least one would contradict the chosen minimum. Therefore its interior is empty.

Key insight: A minimum-area counterexample cannot contain a smaller one.

Hard: A Path Through A Tournament

Choose a longest directed path $v_1 \rightarrow \dots \rightarrow v_k$. Suppose a vertex u is omitted. If $u \rightarrow v_1$, prepend u ; if $v_k \rightarrow u$, append u . Otherwise $v_1 \rightarrow u$ and $u \rightarrow v_k$. Moving along the path, there is an index i with $v_i \rightarrow u$ and $u \rightarrow v_{i+1}$, so u can be inserted between them. Every case contradicts maximality, so the path contains every vertex.

Key insight: A longest object is rigid enough that an omitted vertex causes a contradiction.

Very Hard: Longest Paths Meet

Let two longest paths P and Q both have L edges. Suppose they are disjoint. Since the graph is connected, choose a shortest path joining a vertex p of P to a vertex q of Q , with no other vertices on $P \cup Q$. From p , one endpoint of P is at distance at least $L/2$ along P ; similarly, one endpoint of Q is at distance at least $L/2$ from q . Joining these two subpaths through the connecting path produces a path longer than L , a contradiction.

Key insight: Two disjoint extremes can be joined to create something more extreme.

Stretch: Erdős–Szekeres Monotone Subsequence

For each term a_i , let L_i be the length of the longest increasing subsequence starting at a_i , and D_i the longest decreasing. If $a_i < a_j$ with $i < j$, then $L_i > L_j$; if $a_i > a_j$, then $D_i > D_j$. Thus all pairs (L_i, D_i) are distinct. Each L_i, D_i is between 1 and n (otherwise we have the desired subsequence). There are at most n^2 distinct pairs, but $n^2 + 1$ terms. By pigeonhole, some pair repeats, contradiction unless a subsequence of length $n + 1$ exists.

Key insight: Map each term to a pair of lengths; pigeonhole principle forces a collision unless the bound is reached.

4 Reduction

Basic: An Acquaintance Graph

Represent each person by a vertex and join two vertices when those people are acquainted. Every degree is one of $0, 1, \dots, n - 1$. However, degrees 0 and $n - 1$ cannot both occur: if one person knows everyone, nobody knows zero people. Thus at most $n - 1$ different degree values occur among n vertices. By the pigeonhole principle, two vertices have the same degree.

Key insight: The graph representation turns acquaintances into vertex degrees.

Easy: The Bridges Of Königsberg

Replace each land mass by a vertex and each bridge by an edge. A walk using every bridge exactly once would be an Euler trail. In such a trail, every vertex except possibly the start and finish has even degree. The Königsberg graph has four odd-degree vertices, so no Euler trail exists.

Key insight: Only incidence matters, so geography can be discarded.

Medium: Missionaries And Cannibals

State: (M_L, C_L, B) with M missionaries, C cannibals on left, $B \in \{L, R\}$ boat position. On each bank, safety requires that there are no missionaries or that missionaries are at least as numerous as cannibals. Graph edges are valid crossings with one or two passengers. A BFS from $(3, 3, L)$ to $(0, 0, R)$ gives the following 11 crossings (M = missionary, C

= cannibal):

$$\begin{array}{ccccccc} CC \rightarrow, & C \leftarrow, & CC \rightarrow, & C \leftarrow, & MM \rightarrow, & MC \leftarrow, \\ MM \rightarrow, & C \leftarrow, & CC \rightarrow, & C \leftarrow, & CC \rightarrow. \end{array}$$

Checking the counts on each bank after every crossing shows that none ever leaves a missionary outnumbered.

Key insight: State-space graph with safety constraints; BFS finds valid path.

Hard: Lamps As Linear Algebra

Encode on/off by 1/0 in $\mathbb{F}_2$. Let v be the initial state and let column i of an $n \times n$ matrix A have ones in positions $i - 1, i, i + 1$ modulo n . If x_i records whether move i is used, then reaching all off is equivalent to

$$Ax = v \quad \text{over } \mathbb{F}_2.$$

Gaussian elimination decides whether this system is consistent and, when it is, returns a sequence of moves. Repeating a move is unnecessary because doing it twice has no effect.

Key insight: Toggle operations become vector addition modulo 2.

Very Hard: Assignment As Maximum Flow

Create a source s , one vertex for each student, one for each project, and a sink t . Add capacity-1 edges from s to every student, from a student to each acceptable project, and from every project to t . An integral flow of value n selects one outgoing student–project edge for every student and uses every project at most once. Conversely, any valid assignment gives such a flow. Therefore an assignment exists exactly when the maximum flow has value n .

Key insight: Capacity constraints encode distinct choices, reducing assignment to flow.

Advanced Material

The remaining solutions cover optional extensions. Their theory and worked examples are presented in *Week 2 Advanced Slides*.

5 Advanced Logic And Common Knowledge

Basic: Blue-Eyed Islanders ($k = 2$)

If $k = 1$, the single blue-eyed islander sees no blue eyes, knows the announcement refers to them, and leaves night 1. If $k = 2$, each blue-eyed islander sees one blue-eyed person. If I had brown eyes, the other would see 0 blue eyes and leave night 1. They don't, so I must have blue eyes. Both blue-eyed islanders leave at midnight on day 2. Each brown-eyed islander sees two blue-eyed people; after observing both leave on day 2, they know they are brown. If one of them were blue-eyed, there would have been three blue-eyed islanders, none of whom would leave on day 2. Thus the brown-eyed islanders leave at midnight on day 3.

Key insight: Public announcement creates common knowledge; induction on k gives the day of departure.

Easy: Cheating Husbands

Let k be the number of cheating husbands. By the same induction as Blue-Eyed Islanders, all k wives deduce their husbands cheated on day k and act at midnight. The queen's truthful announcement ensures that $k \geq 1$.

Medium: Sum And Product Puzzle

The information limit $x + y \leq 100$ makes systematic elimination possible. Start with all pairs $\Omega_0 = \{(x, y) : 2 \leq x < y, x + y \leq 100\}$. Keep in Ω_1 only pairs whose product occurs for *at least two* pairs in Ω_0 (P's first statement). Keep in Ω_2 only pairs for which *every* Ω_0 pair with the same sum belongs to Ω_1 (S knew P could not know). The possible sums in Ω_2 are

$$11, 17, 23, 27, 29, 35, 37, 41, 47, 53.$$

Keep in Ω_3 only pairs whose product occurs *once* in Ω_2 (P now knows). Finally keep in Ω_4 only pairs whose sum occurs *once* in Ω_3 (S now knows). Enumerating the pairs under the given bound yields $\Omega_4 = \{(4, 13)\}$. For example, product 52 has factor pairs (2, 26) and (4, 13); only the latter has a sum surviving S's first statement. The answer is $x = 4$, $y = 13$. These filters spell out exactly which set of possibilities each speaker has at every stage.

Key insight: Filter a finite set of possible worlds after each public statement.

Hard: Cheryl's Birthday

Dates: May 15, 16, 19; June 17, 18; July 14, 16; Aug 14, 15, 17. Albert knows month, Bernard knows day. Albert: Bernard doesn't know $\rightarrow$ month is not May or June (19 and

18 are unique days). Remaining: July 14, 16; Aug 14, 15, 17. Bernard now knows $\rightarrow$ day is not 14. Remaining: July 16; Aug 15, 17. Albert now knows $\rightarrow$ month must be July (if August, Albert couldn't distinguish 15 and 17). Answer: July 16.

Very Hard: 100 Prisoners And A Light Bulb

Designate one prisoner as the Counter. Others are Drones. Protocol: A Drone who has never turned the light on, and finds it off, turns it on exactly once. The Counter only turns the light off, and increments a mental count each time they find it on. When the Counter's count reaches 99, they announce. Proof: Because visits are independent and uniform, every prisoner is selected infinitely often with probability 1. Whenever the light is on, the Counter eventually visits, turns it off, and records one signal. Whenever it is off and an unsignalled Drone remains, an unsignalled Drone eventually visits and turns it on. These stages repeat, so all 99 Drones eventually signal almost surely. Each Drone signals at most once, so the count reaches 99 only after every Drone has visited. The announcement is therefore correct, and termination has probability 1.

Key insight: The light bulb encodes one bit of shared state; the Counter aggregates information.

6 Advanced Invariants And Monovariants

Basic: Peg Solitaire (Small Board)

No legal first move exists. The centre is the only empty cell, but an orthogonal jump landing there would have to start two squares away from the centre, outside the 3×3 board. Thus the initial configuration is already stuck with eight pegs.

Key insight: Check whether a first move exists before seeking a more elaborate invariant.

Easy: Token Game

Initially the only adjacent pair is at 0 and 1, so the only move replaces those two tokens with one token at 2. No further move is possible with just one token. Position 100 is never occupied.

Key insight: Check the legal moves before searching for an invariant.

Medium: Conway's Soldiers

Fix a possible target square $(x_0, 5)$ and put $\varphi = (\sqrt{5} - 1)/2$, so $\varphi + \varphi^2 = 1$. Assign cell (x, y) weight $\varphi^{|x-x_0|+|y-5|}$. For any three consecutive cells in a row or column, the weight of the destination is at most the sum of the weights of the two cells from which a peg jumps. In the only tight case, their distances from the target are $k+2, k+1, k$ and the inequality is precisely $\varphi^{k+2} + \varphi^{k+1} = \varphi^k$. Removing the jumped peg therefore never increases total weight.

The initially occupied cells have total weight exactly 1:

$$\sum_{y \leq 0} \sum_{x \in \mathbb{Z}} \varphi^{|x-x_0|+5-y} = \frac{\varphi^5}{1-\varphi} \left(1 + \frac{2\varphi}{1-\varphi} \right) = 1.$$

The target itself has weight 1. After any finite number of jumps, infinitely many initial pegs remain elsewhere, with positive total weight. Reaching the target would give total weight greater than 1, a contradiction. This holds for every x_0 , so row 5 is unreachable.

Medium: Polynomial Invariant For (x+2,y-1)

The move $(x, y) \rightarrow (x+2, y-1)$ changes $x+2y$ by $2+2(-1) = 0$. Thus $P(x, y) = x+2y$ is a polynomial invariant. More generally, any polynomial in $x+2y$ is invariant. This is a *necessary* condition for reachability, not sufficient: the move can only be applied a non-negative integer number k of times. The exact reachable set is $\{(2k, -k) : k \in \mathbb{Z}_{\geq 0}\}$, which lies on the line $x+2y = 0$.

Key insight: An invariant excludes states, while the direction of the move and integrality complete the reachability argument.

Very Hard: A Sequence Invariant

Set $b_n = a_n + 1$. The given recurrence becomes the Fibonacci-type recurrence $b_{n+1} = b_n + b_{n-1}$, with $b_0 = 1$, $b_1 = 2$ and $b_2 = 3$. For $d_n = b_n^2 - b_{n-1}b_{n+1}$,

$$\begin{aligned} d_{n+1} &= b_{n+1}^2 - b_n b_{n+2} \\ &= b_{n+1}(b_n + b_{n-1}) - b_n(b_{n+1} + b_n) \\ &= b_{n+1}b_{n-1} - b_n^2 = -d_n. \end{aligned}$$

Since $d_1 = b_1^2 - b_0 b_2 = 4 - 3 = 1$, we have $d_n = (-1)^{n-1}$, proving the claimed invariant.

7 Colouring And Tiling: Generalisations

Basic: Mutilated Chessboard: Four Corners Removed

Yes. In each of the top and bottom rows, the two corners are missing, leaving six consecutive squares; tile each row with three horizontal dominoes. Each of the other six rows has eight squares and can be tiled with four horizontal dominoes. These $3 + 3 + 6 \cdot 4 = 30$ dominoes cover every remaining square.

Key insight: An explicit tiling is more decisive than an unnecessary colouring.

Easy: Tiling With T-Tetrominoes

The equal letters in this diagram give four T-shaped pieces:

AAAB
CABB
CCDB
CDDD

Each letter appears in four edge-connected cells forming a T. To tile $4 \times (4k)$, divide it into k disjoint 4×4 blocks and repeat this construction in every block.

Hard: Fault-Free Domino Tiling

For a 6×6 board, consider any one of its five internal horizontal grid lines. The number of vertical dominoes crossing it must be even: the part above the line has an even number of cells, and all other dominoes cover two cells there. If the tiling is fault-free, at least two dominoes cross each such line. The same argument applies to its five internal vertical lines. Since each domino crosses at most one grid line, fault-freeness would require at least $2(5 + 5) = 20$ dominoes, but the board holds only 18.

Here is a fault-free 6×8 tiling; each repeated letter is one domino, and the paired cells are adjacent:

AABBCCDE
FFGGHIDE
JKKLHIMM
JNOLPPQQ
RNOSSTTU
RVVWXXU

Every one of the five internal horizontal and seven internal vertical lines crosses a domino, so there is no fault line.

Hard: Triominoes On A Chessboard

Choose row and column indices starting at the removed corner, and colour (i, j) by $(i + j) \bmod 3$. Every horizontal or vertical 1×3 tromino covers one cell of each colour. The intact board has colour counts 21, 22, 21; removing the $(0, 0)$ corner leaves 20, 22, 21, which are not equal. Therefore no tiling exists.

Very Hard: Generalised Mutilated Chessboard

The answer is $m, n \geq 2$ with mn even, together with the two boards 1×2 and 2×1 (whose empty remainder has an empty tiling). Odd area leaves an odd number of squares and cannot be tiled. For $m, n \geq 2$ with even area, rotate the board if necessary so that its number of rows is even. Cross the top row from left to right, snake through columns 2 through n of the remaining rows, then return up column 1. This is a Hamiltonian cycle of the board's squares. Colours alternate around the cycle; deleting two opposite-colour squares leaves two paths, each containing an even number of squares. Pair consecutive squares on each path to tile the remainder.

Finally, on a $1 \times n$ board of even length $n \geq 4$, remove positions 2 and 3. They have opposite colours but leave position 1 isolated, so this board does not *always* admit a tiling. The same holds after rotation. The 1×2 case is the sole exception.

8 Induction And Recursion: From Proof To Algorithm

Basic: Josephus Problem

$J(1) = 1$. Number the people 1 through n and start by removing person 2. For $2k$ people, the first circuit removes every even position; the k odd-numbered survivors form a smaller Josephus circle, giving $J(2k) = 2J(k) - 1$. For $2k + 1$ people, the first circuit removes $2, 4, \dots, 2k$ and then person 1. The remaining k people are $3, 5, \dots, 2k + 1$, giving $J(2k + 1) = 2J(k) + 1$. Thus the recurrence ranges are $J(2k) = 2J(k) - 1$ for $k \geq 1$ and $J(2k + 1) = 2J(k) + 1$ for $k \geq 1$. These recurrences imply by induction that if $n = 2^r + s$ with $0 \leq s < 2^r$, then $J(n) = 2s + 1$: for even and odd n alike, the formula for $J(\lfloor n/2 \rfloor)$ gives the next case. Equivalently, move the leading bit of n 's binary representation to the end.

Easy: Gray Codes

$G_1 = [0, 1]$. Given G_n , construct G_{n+1} by prefixing 0 to G_n and prefixing 1 to reverse of G_n . Proof by induction: consecutive strings in first half differ by induction; boundary between halves is $0w$ followed by $1w$, where w is the last string of G_n , so only the first bit changes. Consecutive strings in the reversed second half also differ by induction, and every $(n + 1)$ -bit string appears exactly once.

Medium: Recursive 15-Puzzle Parity

Read the numbered tiles row by row, omitting the blank, and count inversions I . Let r be the blank's row counted from the bottom, starting at 1. A horizontal move leaves both parities unchanged; a vertical move changes r by 1 and changes inversion parity by 3. Thus $I + r$ has invariant parity. The goal has $I = 0, r = 1$, so a configuration is solvable only if $I + r$ is odd. This condition is also sufficient by the classical completeness theorem for the 15-puzzle: every configuration with the goal's invariant is reachable.¹ Therefore the decision algorithm returns true exactly when $I + r$ is odd. Count inversions recursively using merge sort, counting cross-inversions while merging; then find the blank's row r and test the parity. This takes $O(16 \log 16)$ time and $O(16)$ space.

Hard: Counterfeit Coin With n Weighings

The answer, for $n \geq 2$ and no initially known genuine coin, is $N_n = (3^n - 3)/2$. The following two auxiliary strategies explain how to attain it.

Signed suspects: If each of 3^r suspect coins has a known direction (it can only be heavy or only be light), r weighings suffice. For $r = 0$ there is one suspect. For the induction step put $t = 3^{r-1}$, and, by reversing heavy/light if needed, let at least half of the $3t$ suspects be heavy candidates. If at least $2t$ are heavy candidates, weigh t heavy candidates against t other heavy candidates: balance leaves t suspects, while a tilt selects the t heavy candidates on the heavier pan. Otherwise put $(t + 1)/2$ heavy and $(t - 1)/2$ light candidates on *each* pan. A balance leaves t suspects off the scale; a tilt leaves the heavy candidates on the heavier pan and the light candidates on the lighter pan, again t in all. Each case reduces to the induction hypothesis.

¹W. E. Story, "Note on the '15' Puzzle," *American Journal of Mathematics* 2 (1879), 399–404.

Known genuine coin: With a genuine reference, we can resolve $U_r = (3^r - 1)/2$ unknown suspects in r weighings. For $r = 1$, compare the suspect with the genuine coin. For $r > 1$, let $t = 3^{r-1}$; weigh $(t + 1)/2$ suspects against $(t - 1)/2$ other suspects and one genuine coin. If balanced, U_{r-1} unweighed suspects remain, still with a genuine reference. If tilted, the t weighed suspects have known heavy/light directions and the signed-suspect strategy applies.

Now split $N_n = 3U_{n-1}$ unknown coins into equal groups A, B, C and weigh A against B . If balanced, C contains the counterfeit and A supplies genuine coins, so use the known-genuine strategy. If tilted, a coin in the heavier group might be heavy *or* one in the lighter group might be light. These are $2U_{n-1} = 3^{n-1} - 1$ signed suspects; add one known genuine coin from C as a dummy candidate and apply the signed-suspect strategy. The dummy cannot be the answer. This constructs a strategy in n weighings.

For optimality, suppose the first weighing compares k unknown coins against k others, leaving $N - 2k$ off the scale. A tilt gives $2k$ signed possibilities and only 3^{n-1} remaining outcome sequences; as $2k$ is even, $2k \leq 3^{n-1} - 1$. A balance gives genuine coins but $N - 2k$ unknown ones, so the known-genuine bound gives $N - 2k \leq U_{n-1}$ (the all-balanced outcome cannot distinguish the two directions of an unweighed counterfeit). Hence $N \leq (3^{n-1} - 1) + U_{n-1} = (3^n - 3)/2$.

Very Hard: Hadamard Matrix Construction

$H_1 = [1]$. $H_{2n} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix}$. Proof: Rows of H_{2n} are either (r, r) or $(r, -r)$ for rows r of H_n . Orthogonality follows from orthogonality of H_n rows. By induction, H_{2^k} is Hadamard.

9 Algorithmic Bridges

Basic: A Maze As A Graph

Create one vertex for every open cell. Join two vertices when their cells share a side. Run BFS from the start, storing distance 0 there and assigning each newly discovered neighbour distance one greater than its parent. The finish distance is the minimum number of moves; if it is never discovered, no route exists. BFS is optimal because it visits vertices in non-decreasing distance from the start, so the first path reaching any cell uses the fewest possible edges.

Easy: Greedy Activity Selection

Sort by finish time $f_1 \leq f_2 \leq \dots \leq f_n$. Pick $i_1 = 1$. For $k > 1$, pick the first i_k with $s_{i_k} \geq f_{i_{k-1}}$. For the exchange proof, let g be the earliest-finishing activity and let o be the first activity in an optimal schedule. Since $f_g \leq f_o$, replacing o by g leaves every later activity compatible: each starts at or after f_o , hence at or after f_g . Thus an optimal schedule can start with the greedy choice. Apply the same argument to the remaining activities, proving optimality by induction.

Medium: DP From Induction: Coin Change

Let $f(s)$ be the minimum number of coins needed to make sum s . Set $f(0) = 0$ and, for $1 \leq s \leq n$, initialise $f(s) = \infty$. Use the recurrence below, with $\min \emptyset = \infty$:

$$f(s) = \begin{cases} 0, & s = 0, \\ 1 + \min_{\substack{1 \leq i \leq k \\ c_i \leq s}} f(s - c_i), & s > 0, \end{cases}$$

Tabulate in increasing order of s , updating for each coin $c_i \leq s$ by $f(s) = \min(f(s), 1 + f(s - c_i))$. An unreachable target retains value ∞ . Time is $O(kn)$ and space is $O(n)$.

Hard: Sliding Puzzle State-Space Search

Encode the whole state as a permutation of the five numbered tiles *and* the blank: there are $6! = 720$ such permutations in total. For a fixed goal, exactly 360 satisfy the puzzle's parity condition and are reachable. A transition swaps the blank with a tile sharing an edge. BFS from the goal explores these states, storing distances and predecessors. If the start state is reached, its recorded distance is the minimum number of moves. Following predecessors from the start leads to the goal; if the start is not reached, it is unsolvable. Alternatively, BFS may start from the supplied start state and read the goal's distance; reconstruct the path by following predecessors from the goal back to the start and reversing that sequence.

Very Hard: Greedy Scheduling With Deadlines

Sort jobs by decreasing profit. Only n slots can be occupied, so replace each deadline by $\min(d_i, n)$. Initialise a parent array on slots $0, 1, \dots, n$ with $\text{parent}[t] = t$. Its $\text{find}(t)$ returns the latest free slot at most t ; 0 is a sentinel. For each job, set $t = \text{find}(\min(d_i, n))$. If $t > 0$, schedule the job there and set $\text{parent}[t] = \text{find}(t - 1)$; if $t = 0$, skip it. This directed parent update, with path compression, preserves the *latest-free-slot* meaning (ordinary union-by-rank would not).

For an exchange proof, maintain an optimal schedule whose previously accepted greedy jobs occupy their assigned slots. A previously rejected job cannot appear in this schedule: when it was rejected, all slots before its deadline were already occupied by earlier accepted jobs. For the next accepted job J , let t be its chosen free slot. If J is absent from the optimal schedule, put it at t , replacing the job there if necessary. That job was not processed earlier, so it has profit at most that of J ; if the slot was empty, positive profit makes the schedule strictly better. If J is already scheduled at another slot u , then $u < t$: every free slot up to its deadline after t is occupied by a previously accepted job. Swap J into t and move any occupant of t to u ; the latter's deadline is at least $t > u$, so feasibility is preserved. In either case an optimal schedule agreeing with one more greedy choice exists.